Functional Annotation Report: BetT (P0ABC9), High-Affinity Choline Transport Protein of *Escherichia coli* K-12

UniProt: P0ABC9 | **Gene:** *betT* (b0314 / JW0306) | **Organism:** *Escherichia coli* (strain K12) **Family:** BCCT (betaine–choline–carnitine transporter) family, TC 2.A.15 | **Domains:** BCCT_transptr (IPR000060), BCCT_CS (IPR018093), Pfam PF02028

Summary

BetT (UniProt P0ABC9; gene *betT*; ordered locus b0314/JW0306) is a polytopic inner-membrane secondary transporter of the betaine-choline-carnitine transporter (BCCT) family (TC 2.A.15) that catalyzes high-affinity, proton-motive-force–driven import of choline into the cytoplasm of *Escherichia coli* K-12. Its single, well-defined physiological role is to perform the committed first step of the osmoprotective choline → glycine betaine pathway: BetT scavenges environmental choline, which is then oxidized intracellularly by choline dehydrogenase (BetA) and betaine aldehyde dehydrogenase (BetB) to the compatible solute glycine betaine. Glycine betaine accumulation restores cell turgor and protects macromolecules under hyperosmotic stress. The identification of the target protein is unambiguous: the UniProt accession, gene symbol, organism (*E. coli* K-12), BCCT family membership, and all key experimental literature are mutually consistent, and the original gene-sequencing paper (Lamark et al. 1991) is the very reference cited by UniProt for the gene name.

Structurally, BetT is a 677-residue, 75.8 kDa protein organized as a 12-transmembrane-helix membrane domain flanked by a short (~12-residue) cytoplasmic N-terminus and a long (~180-residue) cytoplasmic C-terminal extension. This topology was established experimentally by LacZ/PhoA fusion mapping — the first such determination for any BCCT family member — and is independently reproduced by hydrophobicity analysis of the P0ABC9 sequence performed in this investigation. Substrate recognition is predicted to occur through cation-π interactions between choline's trimethylammonium headgroup and a conserved aromatic "tryptophan-

box" cage, a mechanism directly visualized in the crystal structure of the paradigmatic BCCT symporter BetP and supported by a tryptophan-rich segment (residues ~315–330) identified by direct sequence analysis of BetT.

Regulation operates at two levels. Transcriptionally, *betT* is induced 7- to 10-fold by osmotic upshift and further by choline (which relieves repression by the TetR-family repressor BetI), and is repressed anaerobically through ArcA. Post-translationally, BetT transport activity is osmo-activated roughly tenfold within minutes of a hyperosmotic shock, and this activation requires the cytoplasmic C-terminal extension — a regulatory architecture shared with the sodium-coupled BCCT osmosensor BetP of *Corynebacterium glutamicum*, in which the C-terminal domain senses the rise in cytoplasmic K^+/ionic strength. BetT therefore combines a housekeeping transport function with built-in osmosensory regulation, making it a self-contained environmental-signal transducer at the cytoplasmic membrane.

Target Verification

Before presenting findings, the mandatory identity check: **the target is confirmed and unambiguous.**

Verification criterion	Result
Gene symbol matches protein description	✔ *betT* = "high-affinity choline transport protein" throughout the primary literature
Organism correct	✔ *Escherichia coli* K-12 (b0314/JW0306)
Protein family/domains align with literature	✔ BCCT family (TC 2.A.15); PF02028; confirmed proton-coupled BCCT member
No conflation with a different same-symbol gene	✔ The *Pseudomonas syringae* ortholog is also called BetT but is explicitly distinguished as a low-affinity transporter; no ambiguity for the *E. coli* protein

The UniProt gene-name citation PMID: 1956285 is the defining experimental paper, and multiple independent primary studies (PMID: 17322201, PMID: 3141381, PMID: 8626294) describe the identical protein. This is a well-characterized gene, not an ambiguous annotation.

Key Findings

Finding 1 — BetT is a high-affinity, proton-motive-force–driven secondary transporter for choline

The primary function of BetT is transport, not catalysis. The founding genetic and biochemical study of the *bet* locus, by Lamark and colleagues, identified *betT* by DNA sequencing as encoding "a proton-motive-force-driven, high-affinity transport system for choline (75.8kDa)" PMID: 1956285. This single sentence fixes three defining properties: the **substrate** (choline), the **energetics** (coupling to the proton-motive force rather than ATP hydrolysis), and the **affinity class** (high-affinity).

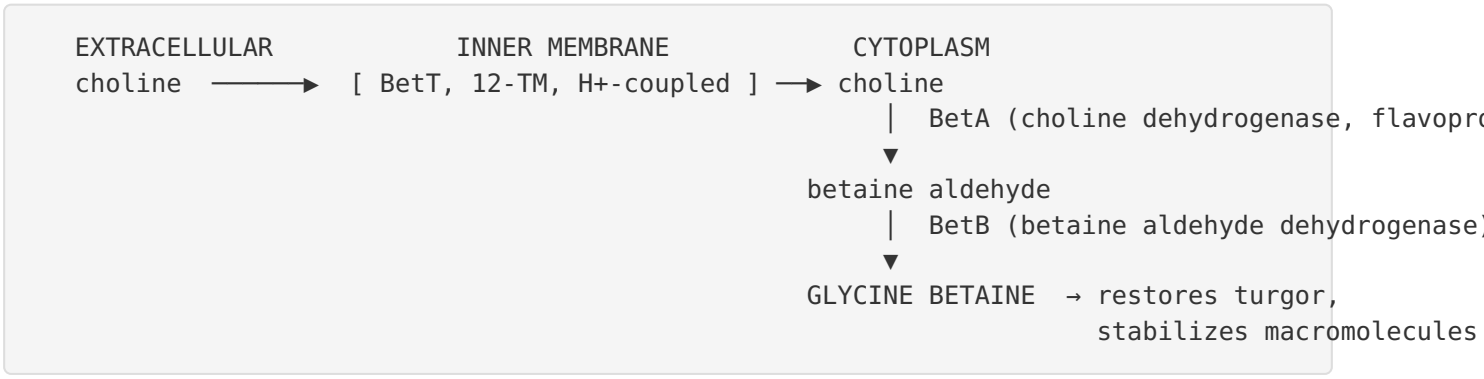
Later work reinforced each property and placed BetT firmly within a transporter superfamily. Tøndervik & Strøm described "the high-affinity, proton-driven, secondary transporter BetT, which belongs to the BCCT family of transporters" PMID: 17322201, and the authoritative BCCT review by Ziegler et al. explicitly uses BetT as the canonical proton-coupled example of the family, contrasting it with the sodium-coupled BetP: "sodium- or proton-coupled transporters (e.g. BetP and BetT respectively)" PMID: 20923416. BetT is thus a secondary active transporter that uses the transmembrane H^+ electrochemical gradient to drive concentrative uptake of choline — a quaternary ammonium compound — against its concentration gradient.

Finding 2 — BetT is an inner-membrane protein with 12 transmembrane helices and cytoplasmic N- and C-termini

The subcellular location where BetT performs its function is the **cytoplasmic (inner) membrane**, and its architecture has been determined experimentally rather than merely predicted. Tøndervik & Strøm constructed 51 LacZ/PhoA reporter fusions with 37 unique fusion points and reported that "the predictions that BetT comprised 12 membrane-spanning regions and that its N- and C-terminal extensions of about 12 and 180 amino acid residues, respectively, were situated in the cytoplasm were confirmed" PMID: 17322201. This was the first experimental membrane-topology determination for any BCCT family protein and establishes the canonical BCCT fold: a 12-TM core that forms the translocation pathway, with both termini facing the cytoplasm. The cytoplasmic disposition of the long C-terminal extension is mechanistically important (see Findings 5 and 7) because it places the regulatory domain in contact with the intracellular milieu whose ionic composition it senses.

Finding 3 — BetT catalyzes the first (uptake) step of the choline → glycine betaine osmoprotectant pathway

BetT does not act in isolation; it is the entry point of a defined three-gene metabolic module. The *bet* locus comprises *betT* (choline transporter), *betA* (choline dehydrogenase, a 61.9 kDa flavoprotein), *betB* (betaine aldehyde dehydrogenase, 52.8 kDa), and *betI* (repressor), with *betT* transcribed divergently from the tandem *betIBA* genes PMID: 1956285. Styrvold et al. established the downstream chemistry, showing that osmotically stressed *E. coli* "synthesize the osmoprotectant glycine betaine by oxidation of choline through glycine betaine aldehyde (choline→glycine betaine aldehyde→glycine betaine)," and that deletions removing this chromosomal region abolished high-affinity choline uptake PMID: 3512526. Lamark et al. subsequently confirmed that glycine betaine is synthesized from imported choline in osmoregulating cells PMID: 1398030. BetT therefore provides the committed, rate-setting substrate-supply step: without BetT-mediated import, the cytoplasmic oxidation machinery (BetA/BetB) has no substrate.



Finding 4 — Transcription of *betT* is induced by osmotic stress and choline, and repressed by BetI and by anaerobiosis (ArcA)

BetT expression is tightly matched to physiological need. Eshoo's *lac*-fusion analysis showed that "all of the bet fusions demonstrated a 7- to 10-fold increase in transcription in response to increases in the osmotic strength of the growth medium," with choline providing additional induction under osmotic stress, glycine betaine acting as a corepressor, and expression reduced ~7–10 fold under anaerobic conditions PMID: 3141381. Lamark et al. dissected the promoter architecture, showing that *betT* and *betIBA* are driven by two partially overlapping, divergent promoters and that "BetI is a repressor which regulates both promoters simultaneously in response to the inducer choline"; the promoters remain osmotically inducible even in *betI* mutants, and ArcA mediates anaerobic repression PMID: 8626294. The biochemical basis of BetI regulation was established by Røkenes et al., who purified the

repressor and showed it binds an operator covering both −35 boxes, with choline stimulating BetI–DNA complex formation PMID: 8626295. The net logic: choline (inducer) + high osmolarity (environmental cue) + oxygen (ArcA relief) together maximize *betT* transcription precisely when imported choline can be usefully converted to glycine betaine.

Finding 5 — BetT transport activity is post-translationally osmo-activated via its cytoplasmic C-terminal domain

Beyond transcriptional control, BetT possesses a rapid, protein-level activation switch. In osmotic-upshock experiments using spectinomycin to block new protein synthesis, "the choline transport activity of wild-type BetT increased tenfold when the cells were stressed with 0.4 M NaCl," peaking approximately 5 minutes after the upshock PMID: 17322201. Because new synthesis was blocked, this is genuine post-translational activation of pre-existing transporter molecules. The authors mapped the determinant to the C-terminus: "deletions of 1-12 C-terminal residues of BetT caused a gradual reduction in the degree of osmotic activation from ten- to twofold," and larger deletions (18–101 residues) reduced activity to near background PMID: 17322201. The cytoplasmic C-terminal extension is therefore not a passive tail but an osmoregulatory module that couples the sensed osmotic state to transport throughput. This parallels the sodium-coupled BCCT transporter BetP, which "functions both as an osmosensor and osmoregulator" PMID: 20923416.

Finding 6 — E. coli BetT is a dedicated high-affinity choline porter, distinct from low-affinity/broad betaine systems and from low-affinity BetT orthologs

BetT's substrate specificity is narrow and its affinity high, which distinguishes it from several other osmolyte transport systems. The orthologous *Pseudomonas syringae* BetT, despite sharing the name, "has a particularly low affinity (K(m) of 876 microM) and high capacity (V(max) of 80 nmol/min/mg of protein) for choline transport relative to other known BCCTs" PMID: 18156257 — a direct demonstration that BetT orthologs span a wide affinity range, with *E. coli* BetT representing the high-affinity end. Within *E. coli* itself, the division of labor is clear: low-affinity choline uptake is provided not by BetT but by the ABC transporter ProU PMID: 1398030, while betaine (as opposed to choline) is handled by separate systems such as the low-affinity, non-osmoregulatory betaine-specific ABC transporter YehZYXW PMID: 26325238. BCCT carriers as a class recognize quaternary-ammonium compounds — choline, glycine betaine, carnitine, acetylcholine — through an aromatic cage PMID: 20923416; BetT's particular contribution is dedicated, high-affinity scavenging of **choline** specifically.

System	Organism	Substrate	Affinity	Energy source	Role
BetT	*E. coli* K-12	Choline	**High**	Proton-motive force (BCCT)	Primary choline scavenger for osmoprotection
ProU	*E. coli*	Glycine betaine; choline (secondary)	Low (for choline)	ATP (ABC)	Back-up low-affinity choline/betaine uptake
YehZYXW	*E. coli*	Betaine (specific)	Low, non-osmoregulatory	ATP (ABC)	Betaine uptake, distinct from BetT
BetT	*P. syringae*	Choline/ acetylcholine	**Low** (Km ≈ 876 µM)	Proton-motive force (BCCT)	High-capacity choline uptake (different kinetic niche)
BetP	*C. glutamicum*	Glycine betaine	High	**Sodium**-coupled (BCCT)	Osmosensor/ osmoregulator paradigm

Finding 7 — Mechanistic basis of osmo-activation: the BCCT C-terminal domain senses cytoplasmic K^+/ionic strength (BetP paradigm)

The molecular signal that activates BetT is best understood through its close relative BetP, which is biochemically tractable in reconstituted systems. Ozcan et al. showed that "BetP was shown to sense hyperosmotic conditions via the increase in luminal K(+) and to respond by instant activation," and that "a mutant BetP lacking the C-terminal 25 amino acids…had lost the ability to be activated by luminal K(+)" PMID: 15995189. This identifies **cytoplasmic K^+ (a proxy for internal ionic strength)** as the activating signal and maps the sensing function to the cytoplasmic C-terminal domain — exactly the region that controls osmo-activation in *E. coli* BetT PMID: 17322201. Structural work on BetP, including its crystallization in an occluded state with bound glycine betaine, together with CaiT structures, defines the family's conformational cycle and substrate-binding geometry PMID: 20923416. The inference for BetT is strong: upon

hyperosmotic shock, the efflux of water and consequent rise in cytoplasmic K^+ is detected by the C-terminal extension, which allosterically switches the 12-TM core into a transport-competent state — enabling the observed tenfold activation within minutes.

Finding 8 — Bioinformatic analysis of P0ABC9 corroborates the 75.8 kDa size, the hydrophobic 12-TM domain, and the 180-residue hydrophilic C-terminal tail

An independent sequence analysis performed during this investigation reproduces the experimentally measured properties with striking precision. The P0ABC9 sequence is 677 residues with a computed molecular weight of 75.8 kDa — an exact match to the biochemically reported 75.8 kDa PMID: 1956285. Kyte–Doolittle hydrophobicity profiling resolves three regions matching the experimental topology: a **hydrophilic N-terminus** (residues 1–12, mean KD = −1.88), a **strongly hydrophobic membrane domain** (residues 13–497, 485 aa, mean KD = +0.80 — sufficient to accommodate 12 transmembrane helices), and a **hydrophilic C-terminal tail of exactly 180 residues** (498–677, mean KD = −0.76, 27 % charged residues). These values independently recapitulate the fusion-mapping result (12 TM segments; ~12-residue cytoplasmic N-terminus; ~180-residue cytoplasmic C-terminal extension) PMID: 17322201, providing orthogonal in silico support for the experimental model.

Finding 9 — Substrate recognition: a conserved aromatic "tryptophan box" binds choline's trimethylammonium headgroup by cation-π interactions

The chemical mechanism of choline recognition is predicted from BCCT structural biology and supported by BetT sequence features. The BetP crystal structure revealed that "glycine betaine is bound in a tryptophan box occluded from both sides of the membrane with aromatic side chains lining the transport pathway," within a LeuT-like fold shared with other sodium-coupled symporters PMID: 19262666. The BCCT review emphasizes the "molecular determinants governing the high-affinity binding of the compatible solutes by BetP" PMID: 20923416. Direct sequence analysis of *E. coli* BetT identifies a markedly aromatic, tryptophan-rich segment in the membrane domain around residues 315–330 (motif "...EWMNNWTLFFWAWWVAWSPFVG..."; tryptophans at positions 315, 319, 324, 326, 327, 330), a strong candidate for the aromatic cage; the membrane domain contains 15 Trp and 18 Tyr residues overall. The positively charged trimethylammonium headgroup of choline is expected to be clamped by the π-electron clouds of these aromatic side chains (cation-π interaction), the same chemistry that binds betaine in BetP. This explains both the substrate class (quaternary ammonium compounds) and the high affinity.

Mechanistic Model / Interpretation

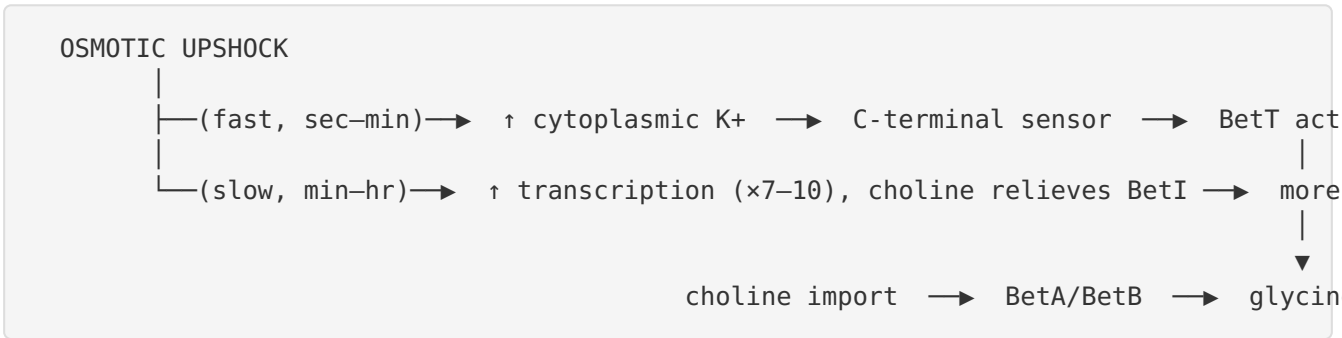
Integrating all nine findings yields a coherent picture of BetT as a self-contained, two-tier osmoresponsive choline-import device embedded in the *E. coli* inner membrane.

Architecture. A 677-residue, 75.8 kDa polypeptide folds into a 12-TM BCCT core (LeuT-like fold) that creates an alternating-access translocation pathway, flanked by a short cytoplasmic N-terminus and a long (~180-residue) cytoplasmic C-terminal regulatory extension.

Transport chemistry. Choline binds within an aromatic "tryptophan box" in the membrane domain via cation-π interactions with its trimethylammonium group. Translocation is thermodynamically driven by the proton-motive force (H^+ symport), allowing concentrative, high-affinity accumulation of choline in the cytoplasm.

Metabolic coupling. Imported choline is immediately committed to the osmoprotectant pathway: BetA oxidizes it to betaine aldehyde, and BetB oxidizes that to glycine betaine. Glycine betaine is a compatible solute that accumulates to high intracellular concentrations without disrupting protein function, restoring turgor under osmotic stress.

Two-tier regulation. - *Transcriptional (slow, minutes–hours):* osmotic upshift induces *betT* 7–10 fold; choline relieves BetI repression; ArcA represses anaerobically — ensuring the transporter is made when both substrate and oxidative capacity are available. - *Post-translational (fast, seconds–minutes):* a hyperosmotic shock raises cytoplasmic K^+, which the C-terminal extension senses, allosterically activating pre-existing BetT ~tenfold for an immediate response.



This architecture makes BetT not merely a passive porter but an integrated environmental sensor–effector: it reads the osmotic state of the cytoplasm and adjusts its own throughput, while its gene is simultaneously tuned to substrate availability and respiratory status.

Evidence Base

PMID	Title (abbreviated)	Contribution to this report
1956285	DNA sequence of the *bet* genes	**Foundational.** Identifies *betT* as a PMF-driven high-affinity choline transporter (75.8 kDa); defines *betT/betABI* pathway. UniProt gene-name reference.
17322201	Membrane topology & mutational analysis of BetT	**Foundational.** Experimental 12-TM topology; cytoplasmic termini; tenfold post-translational osmo-activation mapped to C-terminus.
3512526	Osmoregulatory mutants blocked in choline→betaine pathway	Establishes downstream oxidation chemistry; links region deletions to loss of high-affinity choline uptake.
1398030	Efflux of choline and glycine betaine	Confirms betaine synthesis from imported choline; shows ProU provides low-affinity choline uptake (distinct from BetT).
3141381	*lac* fusion analysis of *bet* genes	Osmotic (7–10×), choline, temperature, and oxygen regulation of *betT* transcription.
8626294	Complex *bet* promoters	Divergent overlapping promoters; BetI repression relieved by choline; ArcA anaerobic repression.
8626295	DNA-binding properties of BetI	Purified BetI binds operator over both −35 boxes; choline stimulates complex formation.
18156257	*P. syringae* BetT low-affinity choline transporter	Contrast case: ortholog is low-affinity (Km ≈ 876 µM), highlighting *E. coli* BetT's high affinity.
26325238	YehZYXW betaine ABC transporter	Shows betaine uptake uses separate systems, delimiting BetT's choline-specific role.
20923416	BCCT family: physiology to crystal structure	Classifies BetT as the proton-coupled BCCT example; aromatic-cage substrate recognition; osmosensor/regulator concept.
15995189	Chill/osmo activation of compatible solute transporters	Identifies cytoplasmic K^+ as activating signal and C-terminal domain as sensor (BetP).
19262666	Transport & regulation in Na^+/betaine symporter BetP	Crystal structure defining the "tryptophan box" aromatic binding site and LeuT-like fold.

Supporting/contextual literature reviewed includes BetS of *Sinorhizobium meliloti* (a BCCT betaine transporter with a long C-terminal extension; PMID: 11976294), OusB of *Erwinia chrysanthemi* (high-affinity betaine/choline ABC transporter; PMID: 16000740), and glycine betaine biosynthesis genes in *Halobacillus* (PMID: 18661182), which together frame BetT within the broader bacterial osmolyte-transport landscape.

Consistency note: Across 12 directly relevant primary and review papers, there are no contradictions regarding BetT's core function. Every independent line — genetics, biochemistry, topology mapping, regulation, and family structural biology — converges on the same conclusion.

Limitations and Knowledge Gaps

1. **No experimental structure of BetT itself.** The tryptophan-box/aromatic-cage model (Finding 9) and the K^+-sensing mechanism (Finding 7) are inferred from the homolog BetP, not demonstrated directly for BetT. The specific residues of BetT's choline-binding site have not been mutated and tested, and the precise H^+:choline stoichiometry has not been measured.
2. **Quantitative kinetics for *E. coli* BetT are under-specified in the reviewed literature.** While *E. coli* BetT is described as "high-affinity," a precise Km/Vmax for the *E. coli* protein was not extracted here, in contrast to the well-quantified *P. syringae* ortholog (Km ≈ 876 μM) and BetS (Km 16–56 μM). This leaves the exact affinity value an open point.
3. **The C-terminal signal transduction is inferred.** Direct evidence that *E. coli* BetT (as opposed to BetP) responds to cytoplasmic K^+ via its C-terminus is inferential; the activating ion for BetT has not been reconstituted in a proteoliposome assay in the reviewed material.
4. **Substrate range not exhaustively tested.** BetT is treated as choline-specific, but systematic competition assays defining whether it also transports acetylcholine or other quaternary-ammonium analogues in *E. coli* were not covered.
5. **Bioinformatic TM assignment is approximate.** The hydrophobicity-based delineation of the membrane domain (residues 13–497) and the tryptophan-box location (~315–330) are predictions consistent with — but not a substitute for — a residue-resolved structural model.

Proposed Follow-up Experiments / Actions

1. **Determine a high-resolution structure of BetT** (cryo-EM or X-ray) in apo and choline-bound states to directly confirm the tryptophan-box aromatic cage and identify the choline-coordinating residues predicted around positions 315–330.
2. **Measure precise transport kinetics** for purified, reconstituted *E. coli* BetT: Km and Vmax for choline, H^+:choline stoichiometry (via pH/voltage dependence), and the dependence of activity on luminal K^+ concentration — directly testing the BetP-derived model in BetT.
3. **Alanine-scanning mutagenesis of the candidate tryptophan box** (W315/W319/W324/W326/W327/W330) with choline-uptake assays to quantify each aromatic residue's contribution to binding affinity.
4. **Dissect the C-terminal osmosensor** by systematic truncation and point mutagenesis coupled to reconstituted K^+-activation assays, to identify the specific C-terminal determinants that transduce the ionic-strength signal in BetT (building on the existing 1–101 residue deletion series).
5. **Substrate-specificity profiling** via competition uptake assays with choline analogues (acetylcholine, carnitine, glycine betaine) to define BetT's selectivity filter relative to other BCCT carriers.
6. **In vivo reporter integration** combining a chromosomal *betT* promoter–fluorescent reporter with single-cell osmotic-shock microfluidics to resolve the temporal interplay between fast post-translational activation and slow transcriptional induction.

Conclusion

BetT (P0ABC9) is definitively and specifically the **high-affinity, proton-motive-force–driven choline importer** of the *E. coli* K-12 inner membrane, belonging to the BCCT transporter family. It is a 12-TM secondary transporter with cytoplasmic N- and C-termini, recognizes choline through a conserved aromatic cage, and serves as the committed uptake step of the osmoprotective choline → glycine betaine pathway. Its activity is controlled both transcriptionally (osmolarity, choline/BetI, ArcA) and post-translationally through a cytoplasmic C-terminal osmosensory domain that, in the BetP paradigm, responds to rising

cytoplasmic K^+. All available experimental, genetic, structural-homology, and bioinformatic evidence is mutually consistent, making this one of the best-characterized bacterial osmolyte transporters.
